

School of Aerospace, Mechanical and Mechatronic Engineering

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Polymer Composite Manufacturing Lab Report

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Name	SID	Contribution
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1 Abstract

This report describes the manufacture and characterisation of a fibre-reinforced polymer (FRP) aerofoil made by manual prepreg lay-up, vacuum bagging and oven curing. Each skin was laid up in an aluminium billet mould as a symmetric four-ply hybrid laminate, with woven carbon/epoxy on both faces and two woven E-glass/epoxy plies at the centre ($[C/G]_s$), and the skins were bonded to a polyurethane foam core. The theoretical overview covers mould making (female and male moulds, process-specific requirements and material selection, mould manufacturing methods, gelcoats and release agents) and the role of each vacuum-bagging consumable. From the measured skin thickness of **[TBC]** mm and the nominal fabric areal masses and fibre densities, the fibre volume fraction was estimated as **[TBC]** %, against about 49 % expected for a void-free laminate; the mould-side surfaces **[TBC: one-line summary of surface quality]**. The discussion compares common FRP manufacturing processes and examines how fibre type, volume fraction, orientation, stacking sequence, voids and cure conditions govern laminate properties, showing that placing the carbon plies outermost roughly doubles the flexural rigidity of the skin. Small UAV wing skins are proposed as a suitable application for hand lay-up.

2 Introduction

2.1 Fibre-Reinforced Polymer Composites

Fibre-reinforced polymer (FRP) composites are multiphase engineering materials comprising reinforcing fibres embedded within a continuous polymer matrix. The fibres provide most of the load-bearing strength and stiffness, while the matrix maintains fibre alignment, transfers inter-fibre stresses and protects the reinforcement from environmental damage [1]. The resulting mechanical properties are generally anisotropic, meaning they depend on loading direction. Through appropriate selection of fibre type, orientation, volume fraction and stacking sequence, FRP laminates can be tailored to specific structural requirements.

A major advantage of FRPs is their high specific strength and stiffness, enabling substantial weight savings in applications where structural efficiency is critical. Unlike metals, which generally exhibit ductile behaviour, or ceramics, which offer high hardness and thermal resistance but are typically brittle, FRPs provide directionally optimised mechanical performance and often excellent corrosion resistance. However, limitations include relatively low through-thickness strength, susceptibility to delamination and temperature restrictions imposed by the polymer matrix [1]. Table 1 summarises the principal engineering differences between these material classes.

Table 1: Qualitative comparison of FRP composites with conventional engineering materials [1].

Property	**FRP Composites**	**Metals**	**Ceramics**
Specific strength and stiffness	Often high; direction-dependent	Moderate to high	Material-dependent
Mechanical behaviour	Anisotropic; may delaminate	Generally ductile	Hard but brittle
Corrosion resistance	Often good	Alloy-dependent	Generally good
Temperature resistance	Limited by matrix	Alloy-dependent	Often excellent
Manufacturing	Lay-up, consolidation and curing	Casting, forming and machining	Forming, sintering and finishing

These characteristics have established FRPs in aerospace, automotive, marine and renewable-energy applications, where lightweight construction and tailored structural performance are particularly valuable [2].

2.2 Fibre Architectures and Matrix Materials

Common reinforcing fibres include carbon, glass and aramid. Carbon fibres offer high specific stiffness and strength, making them suitable for lightweight structural components. Glass fibres provide a cost-effective combination of strength, durability and electrical insulation, while aramid fibres exhibit high toughness and impact resistance [1].

Fibre architecture also strongly influences composite performance. Unidirectional (UD) reinforcement concentrates strength and stiffness along the fibre direction, whereas woven fabrics provide reinforcement along multiple yarn directions. Plain-weave fabrics have frequent interlacements, improving fabric stability, while twill weaves generally offer better drapability over curved surfaces due to reduced interlacement frequency. Satin weaves (e.g. 4- or 8-harness) have even fewer interlacements, giving the best drape and lowest crimp, while non-woven forms such as stitched non-crimp fabrics and randomly oriented chopped-strand mat are chosen for straight, uncrimped fibres and for low cost, respectively [1, 3]. The selection of architecture therefore involves balancing mechanical requirements with manufacturing considerations.

Polymer matrices are broadly classified as thermosetting and thermoplastic. Common thermosets include epoxy, polyester, vinyl ester and phenolic resins, which form cross-linked networks during curing. Thermoplastics, in contrast, soften upon heating and can be reshaped; common examples are polypropylene, polyamides and, for high-temperature aerospace use, polyether ether ketone (PEEK). Epoxy-based prepregs contain fibres pre-impregnated with a controlled resin quantity, improving material consistency and eliminating separate resin application during lay-up. Successful processing requires appropriate consolidation and thermal curing to achieve the intended laminate properties [1, 2].

2.3 Laboratory Objectives

This laboratory investigated the manufacture of a fibre-reinforced polymer composite aerofoil using carbon- and glass-fibre prepregs with a polyurethane foam core. The experiment involved manual prepreg lay-up, vacuum bagging and thermal curing using an aluminium mould [4]. The primary objectives were to:

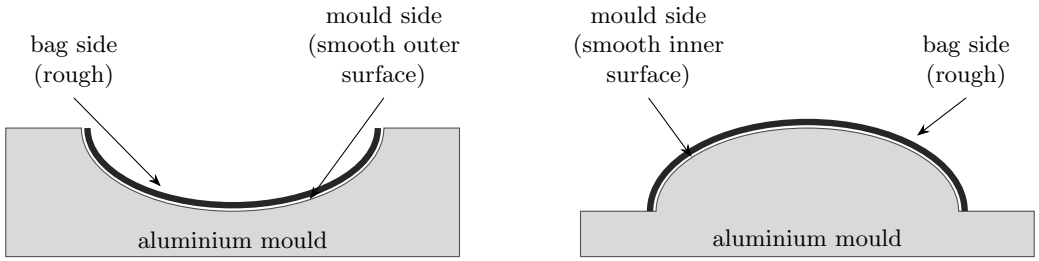
- Understand the key stages of prepreg composite manufacturing, including material preparation, fibre lay-up, vacuum consolidation and curing.
- Examine the effects of fibre type, orientation, stacking sequence and processing conditions on composite quality and structural performance.
- Characterise the manufactured aerofoil through dimensional measurements, laminate architecture, fibre volume fraction estimation and surface inspection.
- Evaluate the advantages, limitations and engineering applications of manual composite lay-up relative to alternative manufacturing techniques.

3 Brief Theoretical Overview

3.1 Mould Design and Manufacture

Making a simple aluminium billet mould. A billet mould like the one used in the laboratory is machined from a solid block of aluminium alloy [3]. (i) *Design:* the tool surface is modelled in CAD from the required outer surface of the part, with a flat margin for trimming, sealant tape and bagging, draft on near-vertical walls, and, for precision parts, an allowance for thermal expansion at the cure temperature. (ii) *Stock preparation:* a plate or billet (e.g. 6061-T6 or stress-relieved cast tooling plate) is cut oversize and faced flat. (iii) *CNC machining:* the cavity is roughed out with flat end mills and finished with ball-nose cutters at a small step-over. (iv) *Finishing:* the cusps are removed by wet sanding with progressively finer abrasives and the surface is polished, since every tool defect is reproduced on the part. (v) *Inspection and preparation:* the profile is checked against the CAD model, and the surface is degreased, sealed and given several cured coats of release agent.

Female versus male moulds. In a female (concave) mould the plies are laid inside a cavity, so the mould-side face becomes the outer surface of the part and the external dimensions are tightly controlled (Figure 1a). This suits parts whose exterior is aerodynamically or cosmetically critical, such as aerofoil skins, boat hulls and body panels, but fabric can bridge across tight internal radii, leaving resin-rich or poorly consolidated corners. In a male (convex) mould the plies are laid over the tool, which controls the inner surface and dimensions (Figure 1b), as needed for ducts or fairings that fit over other structure. Lay-up is easier, but the outer face is the rough bag side, which must be filled and finished if appearance matters, and cure shrinkage makes the part grip the tool, so generous draft is needed [3, 5].



(a) Female (concave) mould (b) Male (convex) mould

Figure 1: Schematic of a laminate (thick line) in (a) a female and (b) a male mould. The mould-side face of the part reproduces the tool finish.

Mould requirements for different processes. Table 2 compares the mould requirements of three processes. Because wet hand lay-up cures near room temperature under contact pressure only, inexpensive moulds of glass-fibre reinforced plastic (GFRP), timber, MDF or plaster suffice. Vacuum infusion also needs a vacuum-tight, stiff tool with a wide flange for sealant tape and the resin and vacuum lines [6]. Prepreg moulds must keep their shape and vacuum integrity through cures at typically 80–180 °C and, in an autoclave, up to about 7 bar [3, 7]. Material selection for prepreg tooling is driven by thermal expansion, thermal mass and cost. Aluminium is light, cheap to machine and heats quickly, but its coefficient of thermal expansion ($\alpha \approx 23 \times 10^{-6}\,\mathrm{K}^{-1}$) far exceeds that of carbon/epoxy ($\alpha \approx 2\text{–}3 \times 10^{-6}\,\mathrm{K}^{-1}$). Heated by 100 K to a typical 120 °C cure, an aluminium tool grows by about 0.23 mm per 100 mm, almost ten times more than the laminate, causing dimensional error and tool–part interaction stresses that can distort thin parts [8]. Steel ($\alpha \approx 12 \times 10^{-6}\,\mathrm{K}^{-1}$) is durable but heavy, while Invar ($\alpha \approx 1.5 \times 10^{-6}\,\mathrm{K}^{-1}$) and carbon/epoxy tooling closely match carbon parts and are preferred for large, tight-tolerance aerospace parts despite their cost [3]. For a small aerofoil like this one, aluminium offers the best balance of cost, machinability and heat-up rate.

Common mould manufacturing methods. (1) Direct CNC machining of metal billet or plate (aluminium, steel, Invar), as used here, gives accurate, durable, high-temperature tools but becomes costly for large parts. (2) Composite (splash) moulds are taken from a master pattern, or plug, that is CNC-machined from tooling board, MDF or foam (or 3D printed), then sealed and polished; tooling gelcoat and a GFRP or CFRP tooling laminate are laid over the plug and stiffened with a backing frame. This is the usual route for large tools such as boat hulls and wind-turbine blades. (3) Direct machining of high-density polyurethane or epoxy tooling board suits prototypes and short runs. (4) Other routes include cast resin or cast metal tools, electroformed nickel shells for very high surface quality and additively manufactured moulds for rapid prototyping [3, 5].

Gelcoats and mould-surfacing chemicals. A gelcoat is a thixotropic, usually pigmented polyester, vinyl ester or epoxy resin applied to the mould as the first layer (typically about 0.5 mm thick). It becomes the outer surface of the part, providing colour and gloss, hiding fibre print-through and protecting the laminate from UV, moisture and abrasion; tooling gelcoats are harder, more heat-resistant grades that form the working surface of composite moulds [5]. Mould sealers fill the microporosity of

Table 2: Typical mould requirements for three composite manufacturing techniques [3, 5–7].

	Wet hand lay-up	**Vacuum infusion**	**Prepreg (vacuum bag/oven or autoclave)**
Cure temperature	Ambient (optional low-temperature post-cure)	Ambient to moderate	Typically 80–180 °C
Pressure	Contact only (brush and roller)	Vacuum, at most ≈1 atm	Vacuum; up to ≈7 bar in an autoclave
Key requirements	Smooth, sealed, releasable surface; low cost	Vacuum-tight and non-porous; stiff under vacuum; wide flange for sealant tape, resin inlet and vacuum line	Stable shape and vacuum integrity at cure temperature; CTE close to the part; low thermal mass; survives many thermal cycles
Typical materials	GFRP with tooling gelcoat; timber, MDF, plaster or foam for one-offs	GFRP or epoxy composite, aluminium, glass or sheet steel	Aluminium, steel, Invar; carbon/epoxy or BMI tooling; high-temperature tooling board
Surface preparation	Wax and/or PVA, or semi-permanent release; part gelcoat sprayed in mould	Semi-permanent release; gelcoat optional	Temperature-stable semi-permanent release; no gelcoat

new or porous tool surfaces so that the release agent forms a continuous film and resin cannot key into the tool. Release agents then prevent adhesion: traditional carnauba waxes (several buffed coats) and water-soluble polyvinyl alcohol (PVA) films, or semi-permanent release agents that bond to the mould, tolerate high cure temperatures and give several releases per application. Mould cleaners remove resin residue and release-agent build-up between uses [9]. In this laboratory only a liquid release agent was wiped onto the bare aluminium moulds (Section 4), as gelcoats are generally not used with prepregs.

3.2 Consumables

The required consumables for composite manufacturing vary between processes. The consumables and their functions for the pre-preg layup performed in the laboratory are as follows:

Mould Release Agent is a coating applied to the mould to prevent the part from bonding to the mould. This ensures that both the part and mould are not damaged during the demoulding process.

A **Breather** is a layer of fabric that is placed on top of the layup and within the vacuum sealed bag. Its primary function is to provide a path for air to travel. Without it, the vacuum bag would suck itself flat to the part, resulting in pockets of air without direct exit paths. The breather creates a distributed pathway from all parts of the bag to ensure that all air is able to escape, producing a proper and effective vacuum.

Peel Ply is another layer of fabric that is laid atop and used to protect the outermost layer of a layup. It prevents the laminate from contamination by dust, oils, and other substances during handling, ensuring that the surface finish of the part is optimal. It also separates the part from the breather and bagging material, so that they can be easily removed. When it is stripped after curing it leaves a clean, uniformly textured surface that is ideal for adhesive bonding; this mattered here because the bag-side surfaces of the skins were later bonded to the foam core [3].

A **Vacuum Bag** is a flexible, airtight film bag. It surrounds and encloses the entire layup so that once air has been fully removed, atmospheric pressure (up to about 101 kPa) compresses the layers, ensuring there are no voids/pockets and that the part is held together during the curing phase.

Tacky Tape is a sealant tape that is lined across the perimeter of the vacuum bag. The bag is pressed onto the tape to create the airtight seal necessary to create a vacuum.

Vacuum Port, Valve and Hose connect the bag to the vacuum pump, with a strip of breather keeping an open air path to the port. More demanding prepreg lay-ups also use release film between the peel ply and breather to control resin bleed, bleeder cloth to absorb excess resin, heat-resistant flash tape, and thermocouples to monitor the part temperature [3, 7]. Figure 2 shows the complete arrangement.

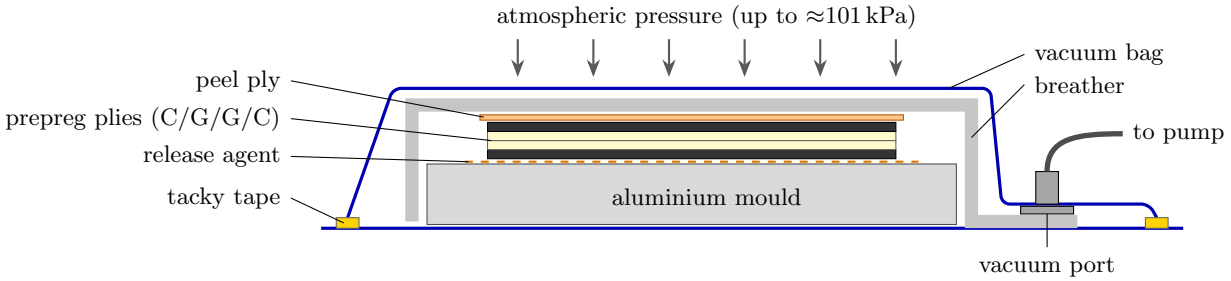


Figure 2: Cross-section (not to scale) of the vacuum-bag arrangement used for each mould: the four prepreg plies on the release-coated aluminium mould are covered by peel ply and breather inside an envelope bag sealed with tacky tape and evacuated through the vacuum port.

4 Lab Procedure

Setup - The following equipment was utilised to carry out the lab procedure: Disposable gloves, ruler, utility knife, female and male mould, pre-preg carbon fibre, pre-preg fibre glass, the consumables covered in section 3, and the vacuum port and line.

The procedure carried out in the lab is as follows:

1 - Measure the mould dimensions using a ruler. Based on this measurement, cut the pre-preg carbon-fibre and fibre-glass to a suitable length and width, ensuring the cut covers the entire mould. There should be a total of four cuts **[TBC: per mould?]**, two carbon-fibre, and two fibre-glass.

2 - Apply the releasing agent to the mould. Only a thin layer is necessary, however, ensure that it is evenly distributed without bare spots. This can be achieved by wiping it across the mould using a cloth.

3 - Lay the carbon-fibre and fibre-glass onto the mould layer by layer. Ensure that each layer is pressed flat against the mould or layer beneath it, minimising the volume of air between layers. The order of the layers should create a symmetrical order about the mid-plane to ensure uniform properties once the part is cured and to stop the laminate warping as it cools from the cure temperature. This is achieved by laying the two carbon-fibre cuts on the top and bottom, sandwiching the two fibre-glass layers. The final result is a 4-ply composite.

4 - Cut a layer of peel ply to the same dimensions as the fibre layups. Place this on top of the topmost layer, once again pressing it flat and minimising any trapped air.

5 - With the main layup finished, place the moulds side by side on top of a vacuum bag. Apply tacky tape to the vacuum bag, creating a perimeter around the moulds.

6 - Cut a single layer of breather fabric large enough to cover both moulds. Place the breather on top of the layup.

7 - Seal the vacuum bag by pulling it back over the moulds. Gradually peel the backing paper of the tacky tape ensuring that the vacuum bag does not wrinkle when pressing it onto the tape,

preventing small channels from emerging. Before finalising the seal, leave one edge of the bag open.

8 - Place the vacuum port at a distance between the mould and bag's edge. The base should sit inside the bag, and screw the top outside the bag, cutting a small slit to allow air to escape. To ensure air has an undisturbed pathway to exit via the port, cut a strip of breather fabric that leads from the primary breather layer to the vacuum port.

9 - Seal the final edge of the vacuum bag. To test the quality of the vacuum, attach the hose to the port and begin the air evacuation process. If there is a noticeable hissing sound, it is highly likely that air is escaping from an unsealed hole around the vacuum bag perimeter.

10 - Once the seal quality is verified, the entire assembly is transferred into the curing oven. The vacuum hose line is connected to a port within the oven. The program should be set for optimum curing cycle recommended by the prepreg manufacturer **[TBC: state the cure cycle used: heating rate, cure temperature and hold time]**. The oven is then started and fan turned on for uniform heating.

11 - Once the curing cycle has been completed and the part has cooled sufficiently, it can be removed from the oven. Once the vacuum is vented, the bag can be cut open and the moulds removed from within. Consumables (e.g. breather, peel ply) can be removed. The part should snap out of the mould with little problem due to the use of the release agent.

12 - The result should be two fibre-reinforced composite parts that make up the upper and lower surface of an aerofoil. These parts are then placed on the outer surface of a foam core with the intended aerofoil profile and bonded using a suitable adhesive **[TBC: name the adhesive used]**.

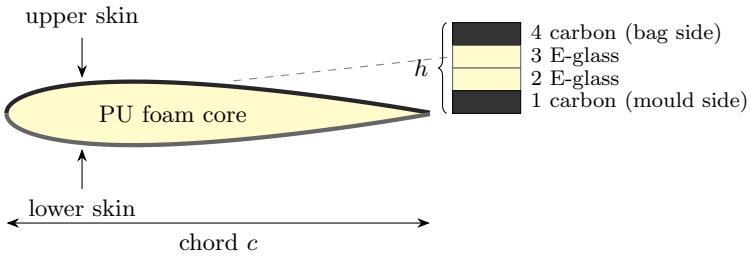
5 Results

5.1 Manufactured Aerofoil

The finished part (Figure 3) is an aerofoil section made of two cured hybrid skins, forming the upper and lower surfaces, bonded to a polyurethane foam core. It has a chord of **[TBC]** mm, a span of **[TBC]** mm, a maximum thickness of **[TBC]** mm ($t/c =$ **[TBC]** %) and a total mass of **[TBC]** g. Skin thickness was measured with **[TBC: a micrometer, ±0.01 mm]** at **[TBC: n]** locations on each skin away from the trimmed edges, lengths with **[TBC: vernier calipers, ±0.02 mm]** and masses with **[TBC: a digital balance, ±0.01 g]**.

[TBC: insert photograph of the finished aerofoil (mould-side and bag-side views)]

(a)



(b)

Figure 3: (a) The manufactured aerofoil. (b) Schematic section (not to scale) showing the two skins bonded to the foam core and the $[C/G]_s$ ply sequence of each skin.

5.2 Laminate Architecture

Each skin is a four-ply, symmetric, inter-ply hybrid laminate with a woven carbon/epoxy ply on each face and two woven E-glass/epoxy plies at the mid-plane (Table 3). With the warp of every fabric

aligned with the span, the lay-up is $[0_C/0_G]_s$, where each 0 denotes a 0/90° woven ply **[TBC: confirm ply angles, fabric styles and prepreg products]**. Carbon and glass each account for half of the plies and, because their nominal areal masses are equal, half of the fibre mass. However, carbon fibre (T300-type, 1.76 g cm^{-3} [10]) is about 30 % less dense than E-glass (2.54 g cm^{-3} [1]), so carbon makes up about 59 % of the fibre volume. The cured epoxy matrix has a density of about 1.2 g cm^{-3} [1].

Table 3: Laminate architecture of each skin (A_w = fabric areal mass, ρ_f = fibre density, W_r = prepreg resin content by mass; matrix density $\rho_m = 1.2\,\mathrm{g\,cm^{-3}}$). Values marked * are nominal values typical of woven epoxy prepregs, used until the supplier data sheets are checked **[TBC: replace with data-sheet values]**.

Ply	Position	Prepreg / fabric	Angle	A_w (g m^{-2})	ρ_f (g cm^{-3})	W_r (wt%)
1	Mould side	Carbon/epoxy, woven **[TBC: weave]**	0/90	200*	1.76	40*
2	Core	E-glass/epoxy, woven **[TBC: weave]**	0/90	200*	2.54	35*
3	Core	E-glass/epoxy, woven	0/90	200*	2.54	35*
4	Bag side	Carbon/epoxy, woven	0/90	200*	1.76	40*

5.3 Fibre Volume Fraction

Because the fabric areal masses are known, the fibre volume fraction can be estimated non-destructively from the cured thickness, using the areal-weight approach of ASTM D3171 Method II [11]:

$$V_f = \frac{1}{h}\sum_i \frac{n_i A_{w,i}}{\rho_{f,i}}, \tag{1}$$

where n_i is the number of plies of fabric i and h the measured laminate thickness. For a void-free laminate with no resin bleed, the cured thickness of each ply follows from the prepreg resin content:

$$t_{\mathrm{ply},i} = \frac{A_{w,i}}{\rho_{f,i}} + \frac{A_{w,i}}{\rho_m}\,\frac{W_{r,i}}{1 - W_{r,i}}. \tag{2}$$

With the values in Table 3, Eq. (2) gives cured ply thicknesses of 0.225 mm for carbon ($V_f = 50.6\,\%$) and 0.168 mm for glass ($V_f = 46.7\,\%$), so the expected skin thickness is $h_{\mathrm{th}} = 0.786$ mm with $V_f \approx 48.9\,\%$ (28.9 % carbon, 20.0 % glass and 51.1 % matrix by volume), an areal mass of 1.28 kg m^{-2} and a density of 1.63 g cm^{-3}. The summed fibre thickness $\sum n_i A_{w,i}/\rho_{f,i}$ is 0.385 mm, so Eq. (1) reduces to $V_f = 0.385/h$ (h in mm), and Table 4 compares the measured skins with this nominal case. Since $V_f \propto 1/h$, a thickness error of only 0.05 mm shifts V_f by about 3 percentage points, so the mean of several readings should be used. If the skin masses and areas are known, comparing the measured density with the void-free value (1.63 g cm^{-3} at the nominal resin content) also gives an estimate of the void content [11].

Table 4: Measured skin thickness and fibre volume fraction from Eq. (1), compared with the nominal void-free laminate.

Quantity	Nominal (void-free)	Upper skin	Lower skin
Thickness h, mean ± s.d. (mm)	0.786	**[TBC]**	**[TBC]**
Mass (g)	–	**[TBC]**	**[TBC]**
Fibre volume fraction V_f (%)	48.9	**[TBC]**	**[TBC]**
Carbon / glass fibre volume (%)	28.9 / 20.0	**[TBC]**	**[TBC]**
Matrix + voids (%)	51.1	**[TBC]**	**[TBC]**

5.4 Surface Quality

[TBC: replace the bracketed observations below with what was seen on your part.] The mould-side (aerodynamic) surfaces were **[TBC: e.g. smooth and glossy, reproducing the**

aluminium finish, with the carbon weave clearly visible]. The bag-side surfaces carried the uniform matte texture imprinted by the peel ply, which gives a clean, keyed surface for bonding to the core. The defects observed were [**TBC: e.g. pinholes or porosity, dry or resin-starved patches, resin-rich areas, wrinkles near the leading or trailing edge, fraying at trimmed edges, gaps or steps at the skin–core joints**], located [**TBC: where**]. After demoulding the skins [**TBC: did / did not**] show warping.

6 Discussion

6.1 Quality of the Manufactured Part

[**TBC: compare the measured thickness and V_f with the nominal values in Table 4.**] A skin thicker than the nominal 0.786 mm (V_f below 49 %) points to trapped air, resin-rich regions or incomplete consolidation, whereas a thinner skin indicates resin bleed into the peel ply and breather. Several features of this process favour some porosity: consolidation relied on vacuum alone (at most $\approx$1 atm), the plies were not debulked individually [**TBC: confirm**], and air trapped between plies of conventional prepreg can escape only through the laminate edges, whereas an autoclave adds several bar of pressure that collapses voids [7, 12]. Manual cutting and placement also cause ply misalignment of a few degrees, wrinkles where the fabric is forced around the tight leading-edge curvature, and bridging across concave regions of the moulds; together with the adhesive bond line, these explain differences between the measured and nominal profile. Quality could be improved by debulking under vacuum after each ply, cutting plies from templates with marked fibre directions, leak-testing the bag before cure, adding a low-temperature dwell to the cure cycle so that air can escape before the resin gels, and, for critical parts, curing in an autoclave [7, 12].

6.2 Comparison of Manufacturing Methods

FRP manufacturing processes trade capital cost, labour and production rate against laminate quality (Table 5). Open-mould processes (wet hand lay-up and spray-up) need the least investment but give low fibre fractions and high, operator-dependent void contents. Vacuum infusion and resin transfer moulding (RTM) make low-void, higher-V_f parts from dry fabric: infusion needs only a single-sided tool and suits very large parts [6], while the matched tools of RTM give two finished surfaces and shorter cycles for medium volumes. Prepreg processes give the most precise control of fibre content and orientation. Autoclave curing, including automated tape laying and fibre placement, gives the highest quality for aerospace primary structure, and the vacuum-bag/oven route used here achieves much of that quality at far lower capital cost, especially with prepregs formulated for out-of-autoclave cure [12]. Filament winding and pultrusion reach high V_f and production rates but are limited to bodies of revolution and constant cross-sections, respectively [1, 3, 5].

6.3 Parameters Influencing Composite Properties

Constituents. Fibre type sets the ceiling on properties: T300-type carbon fibre ($E_f = 230$ GPa, $\rho_f = 1.76\,\mathrm{g\,cm^{-3}}$) is about 4.6 times stiffer per unit mass than E-glass ($E_f \approx 72$ GPa, $\rho_f = 2.54\,\mathrm{g\,cm^{-3}}$) [1, 10], whereas glass is cheaper, has a higher strain to failure and is an electrical insulator. Hybridising them, as here, trades some stiffness for lower cost and better damage tolerance [13]. The matrix governs the matrix-dominated properties (transverse, shear, compressive and interlaminar strength), the maximum service temperature (through its glass-transition temperature) and environmental resistance, while the fibre–matrix interface controls load transfer [1].

Fibre volume fraction. Fibre-direction stiffness follows the rule of mixtures, $E_1 = V_f E_f + (1 - V_f) E_m$ [1], so each extra 1 % of carbon fibre raises the modulus of a unidirectional carbon/epoxy ply by about 2.3 GPa. In practice V_f is limited to about 60–70 % by fibre packing and wet-out, and it is set mainly by the consolidation pressure and resin bleed of the process (Table 5).

Table 5: Comparison of common FRP manufacturing processes. V_f values are indicative only and depend strongly on materials and practice [1, 3, 5, 6].

Process	Principle	V_f (%)	Voids	Tooling / capital cost	Volume; typical applications
Wet hand lay-up	Resin worked into dry fabric by brush and roller in an open mould	20–40	High	Low / low	One-off to low; boat hulls, prototypes, repairs
Spray-up	Chopped roving and resin sprayed onto an open mould	15–30	High	Low / low	Low–medium; baths, tanks, non-structural panels
Prepreg, vacuum bag and oven (this lab)	Prepreg plies laid up, vacuum bagged and oven cured	50–60	Low–mod.	Medium / medium	Low–medium; UAVs, motorsport, sporting goods
Prepreg, autoclave (incl. ATL/AFP)	As above, cured under heat and 3–7 bar	55–65	Very low	High / very high	Low–medium; aerospace primary structure
Vacuum infusion	Resin drawn through a dry preform under a bag by vacuum	50–60	Low	Low / low	Low–medium; wind blades, boat hulls
Resin transfer moulding	Resin injected into a dry preform in a closed matched mould	40–60	Low	High / medium	Medium–high; automotive and aerospace parts
Compression moulding (SMC)	Chopped-fibre charge pressed in heated metal dies	15–35	Low	Very high / high	High; automotive body panels
Filament winding	Resin-wetted tows wound onto a rotating mandrel	55–70	Low	Medium / medium	Medium–high; pressure vessels, pipes, shafts
Pultrusion	Fibres pulled through a resin bath and heated die	50–70	Low	Medium / medium	Continuous; constant-section profiles

Fibre orientation and stacking sequence. Properties are highest along the fibres and fall rapidly off-axis: Krenchel's orientation efficiency factor is 1 for unidirectional fibres, 0.5 for a 0/90° fabric loaded along one fibre direction and 0.375 for random in-plane mat [14]. Woven 0/90° plies therefore give balanced properties in two directions but low in-plane shear stiffness, so ±45° plies are added where torsional stiffness is needed, as in wing skins. A symmetric lay-up removes bending–extension coupling, so the skins do not warp when the carbon and glass plies contract by different amounts on cooling from the cure temperature [1]. Placing the stiffest plies furthest from the mid-plane maximises bending stiffness: a simple lamination estimate using the nominal ply thicknesses and effective ply moduli of about 60 GPa (carbon) and 19 GPa (glass) shows that the $[\mathrm{C/G}]_\mathrm{s}$ skin has about 2.1 times the flexural rigidity of a $[\mathrm{G/C}]_\mathrm{s}$ skin with the same materials and in-plane stiffness.

Void content. Voids act as stress concentrators and crack initiation sites and mainly degrade matrix-dominated properties: interlaminar shear strength falls by roughly 7 % for each 1 % of voids, up to about 4 % void content [15]. Voids come from air trapped during lay-up, moisture and volatiles, and insufficient consolidation pressure, which is why debulking, vacuum integrity and pressure are critical [12].

Processing conditions. The cure cycle (heating rate, dwell temperature and time) controls resin viscosity and flow, the degree of cure and hence the glass-transition temperature, and the exotherm in thick sections, while prepreg that has exceeded its out-life loses tack and flow and becomes more porous [7]. The manufacturing method sets the achievable V_f, void content and fibre straightness (Table 5), and manual operations add ply misalignment and wrinkles, so nominally identical laminates can differ noticeably.

6.4 Application of Hand Lay-up

Hand lay-up suits the wing skins of small fixed-wing unmanned aerial vehicles (UAVs) built in batches

of a few to a few hundred by start-ups, research groups or student teams. Such volumes cannot justify the matched tooling of RTM or an autoclave, whereas hand lay-up needs only a single-sided mould (machined aluminium, tooling board or even 3D printed), a vacuum pump and an oven. Female moulds give the smooth, accurate outer surface needed for low drag, as this laboratory showed. Because every ply is placed by hand, the laminate can be tailored locally, with unidirectional carbon spar caps for bending, $\pm 45^\circ$ plies for torsion and extra plies at attachment points, and glass can replace carbon around antennas because, unlike carbon, it does not block radio signals. Design changes need only a new mould or ply schedule, allowing rapid iteration, and the labour cost and operator-dependent quality are acceptable at these volumes, particularly when prepreg and vacuum bagging are used as here.

7 Conclusion

An FRP aerofoil was made by hand lay-up of woven carbon and E-glass epoxy prepregs in aluminium billet moulds, vacuum bagging and oven curing, and the skins were bonded to a polyurethane foam core.

- Each skin is a symmetric four-ply hybrid $[\mathrm{C}/\mathrm{G}]_\mathrm{s}$ laminate in which carbon is half of the fibre mass but, being less dense (1.76 versus $2.54\,\mathrm{g\,cm^{-3}}$), about 59 % of the fibre volume.
- The fibre volume fraction from the measured thickness was **[TBC]** %, against about 49 % for a void-free laminate, indicating **[TBC: voids / resin bleed]**. The mould-side surfaces **[TBC: summary]**, and **[TBC: main defects]**.
- Moulds must suit the process: aluminium is economical and heats quickly, but expands almost ten times more than carbon/epoxy, so Invar or composite tooling is preferred for large precision parts.
- Fibre type, volume fraction, orientation, stacking sequence, void content and cure govern laminate properties. Placing carbon outermost roughly doubles the skin's flexural rigidity, and the symmetric lay-up prevents warping.
- Hand lay-up remains attractive for low-volume, high-value parts such as UAV wing skins; ply-by-ply debulking or autoclave pressure would further reduce porosity.

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8 Appendix

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